CAO THANH SON

Agentic AI Research Scientist
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PERSONAL PROFILE

PhD in Robotics and AI focused on LLM-based planning for physical robots. I design cognitive architectures that connect language, vision and action in closed-loop systems, then turn the research into production software.

CAREER PATH

Senior Research Scientist, Agentic AI *Mar 2023 – Present*
Orbit Humanoids Lab, Hanoi

- Designed the cognitive architecture for a humanoid robot with hierarchical goal decomposition and failure recovery.
- Built working, episodic and semantic memory, raising multi-step task success by 27%.
- Grounded language reasoning into perception signals and robot control actions in closed-loop trials.
- Defined autonomy metrics for robustness, latency, reliability and safety.

Research Intern, Robot Learning *Jun 2021 – Dec 2021*
Pacific AI Research, Seoul

- Explored multimodal agents that combine vision, language and robot state.

KEY STRENGTHS

Agent systems: Planner-Executor loops, ReAct agents, multi-agent systems, goal decomposition, task orchestration
LLM reasoning: Instruction tuning, safety alignment, hallucination mitigation, RLHF, controllable generation
Memory: Episodic and semantic memory, knowledge graphs, retrieval-augmented reasoning
Engineering: Python, PyTorch, ROS 2, Isaac Sim

SELECTED PUBLICATIONS

- First-author papers at ICRA and NeurIPS on memory-augmented planning for mobile manipulators.
- Co-author of an RSS paper on recovery from failed robot skills.

TRAINING

PhD in Robotics and Artificial Intelligence, KAIST, 2018 – 2023
B.Sc. in Computer Science, University of Science, VNU Hanoi, 2014 – 2018

LANGUAGES

Vietnamese (native), English (fluent), Korean (basic)